```
mysql>       CREATE DATABASE;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '' at line 1
mysql> CREATE DATABASE PROJECT;
Query OK, 1 row affected (0.352 sec)

mysql> USE project;
Database changed
mysql> CREATE TABLE COLLECTION_UNIT (
    ->     unit_id     INT PRIMARY KEY AUTO_INCREMENT,
    ->     unit_name   VARCHAR(100) NOT NULL UNIQUE
    -> );
Query OK, 0 rows affected (0.765 sec)

mysql>
mysql> -- PERSON is the shared-identity superclass resolving the LENDER / employee
Query OK, 0 rows affected (0.005 sec)

mysql> -- overlap from Task 5 (a curator may also be a private lender).
Query OK, 0 rows affected (0.004 sec)

mysql> CREATE TABLE PERSON (
    ->     person_id   INT PRIMARY KEY AUTO_INCREMENT,
    ->     full_name   VARCHAR(150) NOT NULL,
    ->     email       VARCHAR(150) UNIQUE,
    ->     address     VARCHAR(255)
    -> );
Query OK, 0 rows affected (0.441 sec)

mysql>
mysql> -- Multivalued attribute (phone numbers) mapped to its own relation, not CSV.
Query OK, 0 rows affected (0.004 sec)

mysql> CREATE TABLE PERSON_PHONE (
    ->     person_id     INT NOT NULL,
    ->     phone_number  VARCHAR(30) NOT NULL,
    ->     phone_type    ENUM('mobile','work','home') NOT NULL DEFAULT 'mobile',
    ->     PRIMARY KEY (person_id, phone_number),
    ->     FOREIGN KEY (person_id) REFERENCES PERSON(person_id) ON DELETE CASCADE
```

```
    ->     hire_date   DATE NOT NULL,
    ->     unit_id     INT,
    ->     FOREIGN KEY (person_id) REFERENCES PERSON(person_id) ON DELETE CASCADE,
    ->     FOREIGN KEY (unit_id)   REFERENCES COLLECTION_UNIT(unit_id)
    -> );
Query OK, 0 rows affected (0.486 sec)

mysql>
mysql> CREATE TABLE ARTIFACT (
    ->     artifact_id          INT PRIMARY KEY AUTO_INCREMENT,
    ->     catalog_description  VARCHAR(255) NOT NULL,
    ->     unit_id              INT NOT NULL,
    ->     current_lender_id    INT NOT NULL,
    ->     acquisition_date     DATE,
    ->     FOREIGN KEY (unit_id)           REFERENCES COLLECTION_UNIT(unit_id),
    ->     FOREIGN KEY (current_lender_id) REFERENCES LENDER(person_id)
    -> );
Query OK, 0 rows affected (0.731 sec)

mysql>
mysql> -- ------------------------------------------------------------
Query OK, 0 rows affected (0.006 sec)

mysql> -- Ownership history, kept distinct from custody (Changed Requirement, Task 12)
Query OK, 0 rows affected (0.003 sec)

mysql> -- ------------------------------------------------------------
Query OK, 0 rows affected (0.005 sec)

mysql> CREATE TABLE ARTIFACT_OWNER (
    ->     artifact_id       INT NOT NULL,
    ->     owner_person_id   INT NOT NULL,
    ->     ownership_share   DECIMAL(5,4) NOT NULL,
    ->     share_start_date  DATE NOT NULL,
    ->     share_end_date    DATE NULL,
    ->     PRIMARY KEY (artifact_id, owner_person_id, share_start_date),
    ->     FOREIGN KEY (artifact_id)     REFERENCES ARTIFACT(artifact_id),
    ->     FOREIGN KEY (owner_person_id) REFERENCES PERSON(person_id),
    ->     CHECK (ownership_share > 0 AND ownership_share <= 1),
    ->     CHECK (share_end_date IS NULL OR share_end_date > share_start_date)
    -> );
Query OK, 0 rows affected (0.513 sec)

mysql>
mysql> -- ------------------------------------------------------------
Query OK, 0 rows affected (0.003 sec)

mysql> -- Loan coordination core
Query OK, 0 rows affected (0.002 sec)
```

```
    ->     loan_id               INT PRIMARY KEY AUTO_INCREMENT,
    ->     lender_id             INT NOT NULL,
    ->     registrar_id          INT NOT NULL,
    ->     loan_start_date       DATE NOT NULL,
    ->     expected_return_date  DATE NOT NULL,
    ->     actual_return_date    DATE NULL,
    ->     loan_status           ENUM('OPEN','RETURNED','OVERDUE') NOT NULL DEFAULT 'OPEN',
    ->     FOREIGN KEY (lender_id)    REFERENCES LENDER(person_id),
    ->     FOREIGN KEY (registrar_id) REFERENCES EMPLOYEE(person_id),
    ->     CHECK (expected_return_date > loan_start_date),
    ->     CHECK (actual_return_date IS NULL OR actual_return_date >= loan_start_date)
    -> );
Query OK, 0 rows affected (0.589 sec)

mysql>
mysql> -- M:N: one loan carries several artifacts
Query OK, 0 rows affected (0.003 sec)

mysql> CREATE TABLE LOAN_OBJECT (
    ->     loan_id               INT NOT NULL,
    ->     artifact_id           INT NOT NULL,
    ->     condition_at_loan_out  VARCHAR(50),
    ->     PRIMARY KEY (loan_id, artifact_id),
    ->     FOREIGN KEY (loan_id)     REFERENCES LOAN(loan_id),
    ->     FOREIGN KEY (artifact_id) REFERENCES ARTIFACT(artifact_id)
    -> );
Query OK, 0 rows affected (0.449 sec)

mysql>
mysql> -- Custody: physical possession, independent of both LOAN and ARTIFACT_OWNER
Query OK, 0 rows affected (0.004 sec)

mysql> CREATE TABLE CUSTODY (
    ->     custody_id             INT PRIMARY KEY AUTO_INCREMENT,
    ->     artifact_id            INT NOT NULL,
    ->     loan_id                INT NULL,
    ->     custodian_employee_id  INT NOT NULL,
    ->     custody_start_date     DATE NOT NULL,
    ->     custody_end_date       DATE NULL,
    ->     FOREIGN KEY (artifact_id)          REFERENCES ARTIFACT(artifact_id),
    ->     FOREIGN KEY (loan_id)              REFERENCES LOAN(loan_id),
    ->     FOREIGN KEY (custodian_employee_id) REFERENCES EMPLOYEE(person_id),
    ->     CHECK (custody_end_date IS NULL OR custody_end_date > custody_start_date)
    -> );
Query OK, 0 rows affected (0.737 sec)

mysql>
mysql> -- ------------------------------------------------------
Query OK, 0 rows affected (0.004 sec)
```

```
mysql> DELIMITER $$
mysql> CREATE TRIGGER trg_custody_no_overlap
    -> BEFORE INSERT ON CUSTODY
    -> FOR EACH ROW
    -> BEGIN
    ->     DECLARE overlap_count INT;
    ->     SELECT COUNT(*) INTO overlap_count
    ->     FROM CUSTODY
    ->     WHERE artifact_id = NEW.artifact_id
    ->       AND (custody_end_date IS NULL OR custody_end_date > NEW.custody_start_date)
    ->       AND (NEW.custody_end_date IS NULL OR NEW.custody_end_date > custody_start_date)
    ->     IF overlap_count > 0 THEN
    ->         SIGNAL SQLSTATE '45000'
    ->             SET MESSAGE_TEXT = 'Overlapping custody period for this artifact';
    ->     END IF;
    -> END$$
Query OK, 0 rows affected (0.120 sec)

mysql> DELIMITER ;
mysql>
mysql> -- Helpful indexes for the Task 9 workload (overdue loans, missing condition evidence
Query OK, 0 rows affected (0.003 sec)

mysql> CREATE INDEX idx_loan_status      ON LOAN(loan_status, expected_return_date);
Query OK, 0 rows affected (0.412 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> CREATE INDEX idx_condition_art    ON CONDITION_REPORT(artifact_id);
Query OK, 0 rows affected (0.372 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> CREATE INDEX idx_custody_artifact ON CUSTODY(artifact_id, custody_end_date);
Query OK, 0 rows affected (0.506 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM project;
ERROR 1146 (42S02): Table 'project.project' doesn't exist
mysql> SELECT * FROM collection_unit;
Empty set (0.015 sec)

mysql> desc COLLECTION_UNIT;
+-----------+-------------+------+-----+---------+----------------+
| Field     | Type        | Null | Key | Default | Extra          |
+-----------+-------------+------+-----+---------+----------------+
| unit_id   | int         | NO   | PRI | NULL    | auto_increment |
| unit_name | varchar(100) | NO   | UNI | NULL    |                |
+-----------+-------------+------+-----+---------+----------------+
2 rows in set (0.185 sec)
```

```
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql>
mysql> -- ========================================
Query OK, 0 rows affected (0.005 sec)

mysql> -- CSA05 DBMS ASSIGNMENT CO1
Query OK, 0 rows affected (0.004 sec)

mysql> -- MERIDIAN MUSEUM DATABASE
Query OK, 0 rows affected (0.003 sec)

mysql> -- ========================================
Query OK, 0 rows affected (0.003 sec)

mysql>
mysql> -- 1. CREATE DATABASE
Query OK, 0 rows affected (0.003 sec)

mysql>
mysql> DROP DATABASE IF EXISTS MeridianMuseum;
Query OK, 0 rows affected, 1 warning (0.206 sec)

mysql> CREATE DATABASE MeridianMuseum;
Query OK, 1 row affected (0.155 sec)

mysql> USE MeridianMuseum;
Database changed
mysql>
mysql>
mysql> -- 2. CREATE PERSON TABLE
Query OK, 0 rows affected (0.007 sec)
```

```
mysql>
mysql> CREATE TABLE PERSON (
    ->     Person_ID INT PRIMARY KEY AUTO_INCREMENT,
    ->     First_Name VARCHAR(50) NOT NULL,
    ->     Last_Name VARCHAR(50) NOT NULL,
    ->     Email VARCHAR(100) UNIQUE
    -> );
Query OK, 0 rows affected (0.640 sec)

mysql>
mysql>
mysql> -- 3. CREATE EMPLOYEE TABLE
Query OK, 0 rows affected (0.005 sec)

mysql>
mysql> CREATE TABLE EMPLOYEE (
    ->     Employee_ID INT PRIMARY KEY,
    ->     Role VARCHAR(30) NOT NULL,
    ->     FOREIGN KEY (Employee_ID)
    ->     REFERENCES PERSON(Person_ID)
    -> );
Query OK, 0 rows affected (0.348 sec)

mysql>
mysql>
mysql> -- 4. CREATE LENDER TABLE
Query OK, 0 rows affected (0.004 sec)

mysql>
mysql> CREATE TABLE LENDER (
    ->     Lender_ID INT PRIMARY KEY,
    ->     Lender_Type VARCHAR(30) NOT NULL,
    ->     FOREIGN KEY (Lender_ID)
    ->     REFERENCES PERSON(Person_ID)
    -> );
Query OK, 0 rows affected (0.317 sec)

mysql>
mysql>
mysql> -- 5. CREATE CURATOR TABLE
Query OK, 0 rows affected (0.003 sec)
```

```
mysql>
mysql> CREATE TABLE CURATOR (
    ->     Curator_ID INT PRIMARY KEY,
    ->     Department VARCHAR(50),
    ->     FOREIGN KEY (Curator_ID)
    ->     REFERENCES EMPLOYEE(Employee_ID)
    -> );
Query OK, 0 rows affected (0.312 sec)

mysql>
mysql>
mysql> -- 6. CREATE REGISTRAR TABLE
Query OK, 0 rows affected (0.003 sec)

mysql>
mysql> CREATE TABLE REGISTRAR (
    ->     Registrar_ID INT PRIMARY KEY,
    ->     FOREIGN KEY (Registrar_ID)
    ->     REFERENCES EMPLOYEE(Employee_ID)
    -> );
Query OK, 0 rows affected (0.335 sec)

mysql>
mysql>
mysql> -- 7. CREATE ARTIFACT TABLE
Query OK, 0 rows affected (0.002 sec)

mysql>
mysql> CREATE TABLE ARTIFACT (
    ->     Artifact_ID INT PRIMARY KEY AUTO_INCREMENT,
    ->     Artifact_Name VARCHAR(100) NOT NULL,
    ->     Collection_Type VARCHAR(50),
    ->     Description VARCHAR(255)
    -> );
Query OK, 0 rows affected (0.276 sec)

mysql>
mysql>
mysql> -- 8. CREATE LOAN TABLE
Query OK, 0 rows affected (0.002 sec)

mysql>
mysql> CREATE TABLE LOAN (
```

```
mysql>
mysql> CREATE TABLE LOAN (
    ->     Loan_ID INT PRIMARY KEY AUTO_INCREMENT,
    ->     Lender_ID INT NOT NULL,
    ->     Curator_ID INT NOT NULL,
    ->     Start_Date DATE NOT NULL,
    ->     Expiry_Date DATE NOT NULL,
    ->     Return_Date DATE,
    ->     FOREIGN KEY (Lender_ID)
    ->     REFERENCES LENDER(Lender_ID),
    ->     FOREIGN KEY (Curator_ID)
    ->     REFERENCES CURATOR(Curator_ID),
    ->     CHECK (Expiry_Date >= Start_Date),
    ->     CHECK (Return_Date IS NULL OR Return_Date >= Start_Date)
    -> );
Query OK, 0 rows affected (0.826 sec)

mysql>
mysql>
mysql> -- 9. CREATE LOAN_OBJECT TABLE
Query OK, 0 rows affected (0.004 sec)

mysql>
mysql> CREATE TABLE LOAN_OBJECT (
    ->     Loan_ID INT,
    ->     Artifact_ID INT,
    ->     Loan_Value DECIMAL(12,2),
    ->     PRIMARY KEY (Loan_ID, Artifact_ID),
    ->     FOREIGN KEY (Loan_ID)
    ->     REFERENCES LOAN(Loan_ID),
    ->     FOREIGN KEY (Artifact_ID)
    ->     REFERENCES ARTIFACT(Artifact_ID)
    -> );
Query OK, 0 rows affected (0.600 sec)

mysql>
mysql>
mysql> -- 10. CREATE EXHIBITION TABLE
Query OK, 0 rows affected (0.004 sec)

mysql>
mysql> CREATE TABLE EXHIBITION (
    ->     Exhibition_ID INT PRIMARY KEY AUTO_INCREMENT,
```

```
mysql> CREATE TABLE EXHIBITION (
    ->     Exhibition_ID INT PRIMARY KEY AUTO_INCREMENT,
    ->     Exhibition_Name VARCHAR(100) NOT NULL,
    ->     Start_Date DATE,
    ->     End_Date DATE,
    ->     CHECK (End_Date IS NULL OR End_Date >= Start_Date)
    -> );
Query OK, 0 rows affected (0.356 sec)

mysql>
mysql>
mysql> -- 11. CREATE DISPLAY_SLOT TABLE
Query OK, 0 rows affected (0.003 sec)

mysql>
mysql> CREATE TABLE DISPLAY_SLOT (
    ->     Exhibition_ID INT,
    ->     Slot_No INT,
    ->     Slot_Name VARCHAR(50),
    ->     PRIMARY KEY (Exhibition_ID, Slot_No),
    ->     FOREIGN KEY (Exhibition_ID)
    ->     REFERENCES EXHIBITION(Exhibition_ID)
    -> );
Query OK, 0 rows affected (0.330 sec)

mysql>
mysql>
mysql> -- 12. CREATE PLACEMENT TABLE
Query OK, 0 rows affected (0.004 sec)

mysql>
mysql> CREATE TABLE PLACEMENT (
    ->     Loan_ID INT,
    ->     Artifact_ID INT,
    ->     Exhibition_ID INT,
    ->     Slot_No INT,
    ->     Display_Start DATE NOT NULL,
    ->     Display_End DATE,
    ->     PRIMARY KEY (
    ->         Loan_ID, Artifact_ID,
    ->         Exhibition_ID, Slot_No
    ->     ),
```

```
    ->     FOREIGN KEY (Loan_ID, Artifact_ID)
    ->     REFERENCES LOAN_OBJECT(Loan_ID, Artifact_ID),
    ->     FOREIGN KEY (Exhibition_ID, Slot_No)
    ->     REFERENCES DISPLAY_SLOT(Exhibition_ID, Slot_No),
    ->     CHECK (
    ->         Display_End IS NULL
    ->         OR Display_End >= Display_Start
    ->     )
    -> );
Query OK, 0 rows affected (0.594 sec)

mysql>
mysql>
mysql> -- 13. CREATE CONDITION_REPORT TABLE
Query OK, 0 rows affected (0.004 sec)

mysql>
mysql> CREATE TABLE CONDITION_REPORT (
    ->     Report_ID INT PRIMARY KEY AUTO_INCREMENT,
    ->     Loan_ID INT NOT NULL,
    ->     Artifact_ID INT NOT NULL,
    ->     Report_Date DATE NOT NULL,
    ->     Condition_Status VARCHAR(50),
    ->     Remarks VARCHAR(255),
    ->     FOREIGN KEY (Loan_ID, Artifact_ID)
    ->     REFERENCES LOAN_OBJECT(Loan_ID, Artifact_ID)
    -> );
Query OK, 0 rows affected (0.631 sec)

mysql>
mysql>
mysql> -- 14. CREATE ARTIFACT_OWNER TABLE
Query OK, 0 rows affected (0.005 sec)

mysql>
mysql> CREATE TABLE ARTIFACT_OWNER (
    ->     Artifact_ID INT,
    ->     Owner_ID INT,
    ->     Ownership_Share DECIMAL(5,2) NOT NULL,
    ->     Valid_From DATE NOT NULL,
    ->     Valid_To DATE,
    ->     PRIMARY KEY (Artifact_ID, Owner_ID, Valid_From),
    ->     FOREIGN KEY (Artifact_ID)
```

```
   ->     FOREIGN KEY (Artifact_ID)
   ->     REFERENCES ARTIFACT(Artifact_ID),
   ->     FOREIGN KEY (Owner_ID)
   ->     REFERENCES LENDER(Lender_ID),
   ->     CHECK (
   ->         Ownership_Share > 0
   ->         AND Ownership_Share <= 100
   ->     ),
   ->     CHECK (
   ->         Valid_To IS NULL
   ->         OR Valid_To >= Valid_From
   ->     )
   -> );
uery OK, 0 rows affected (0.537 sec)

ysql>
ysql>
ysql> -- 15. CREATE LENDER_CONTACT TABLE
uery OK, 0 rows affected (0.004 sec)

ysql>
ysql> CREATE TABLE LENDER_CONTACT (
   ->     Contact_ID INT PRIMARY KEY AUTO_INCREMENT,
   ->     Lender_ID INT NOT NULL,
   ->     Phone_Number VARCHAR(20),
   ->     FOREIGN KEY (Lender_ID)
   ->     REFERENCES LENDER(Lender_ID)
   -> );
uery OK, 0 rows affected (0.507 sec)

ysql>
ysql>
ysql> -- ========================================
uery OK, 0 rows affected (0.005 sec)

ysql> -- INSERT SAMPLE RECORDS
uery OK, 0 rows affected (0.003 sec)

ysql> -- ========================================
uery OK, 0 rows affected (0.004 sec)

ysql>
ysql> -- PERSON
```

```
mysql>
mysql> INSERT INTO PERSON VALUES
    -> (1,'Arun','Kumar','arun@gmail.com'),
    -> (2,'Ravi','Raj','ravi@gmail.com'),
    -> (3,'Ashok','Kumar','ashok@gmail.com'),
    -> (4,'Vijay','Kumar','vijay@gmail.com'),
    -> (5,'Mani','Raja','mani@gmail.com');
Query OK, 5 rows affected (0.087 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql>
mysql>
mysql> -- EMPLOYEE
Query OK, 0 rows affected (0.004 sec)

mysql>
mysql> INSERT INTO EMPLOYEE VALUES
    -> (1,'Curator'),
    -> (2,'Registrar');
Query OK, 2 rows affected (0.075 sec)
Records: 2  Duplicates: 0  Warnings: 0

mysql>
mysql>
mysql> -- LENDER
Query OK, 0 rows affected (0.004 sec)

mysql>
mysql> INSERT INTO LENDER VALUES
    -> (3,'Individual'),
    -> (4,'Individual'),
    -> (5,'Individual');
Query OK, 3 rows affected (0.082 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql>
mysql>
mysql> -- CURATOR
Query OK, 0 rows affected (0.005 sec)

mysql>
mysql> INSERT INTO CURATOR VALUES
```

```
mysql> INSERT INTO LOAN_OBJECT VALUES
    -> (1,1,50000),
    -> (1,2,75000),
    -> (2,3,60000),
    -> (2,4,45000),
    -> (3,5,30000);
Query OK, 5 rows affected (0.073 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql>
mysql>
mysql> -- EXHIBITION
Query OK, 0 rows affected (0.005 sec)

mysql>
mysql> INSERT INTO EXHIBITION
    -> (Exhibition_Name,Start_Date,End_Date)
    -> VALUES
    -> ('History Exhibition','2026-02-01','2026-11-30'),
    -> ('Art Exhibition','2026-03-01','2026-12-15');
Query OK, 2 rows affected (0.064 sec)
Records: 2  Duplicates: 0  Warnings: 0

mysql>
mysql>
mysql> -- DISPLAY_SLOT
Query OK, 0 rows affected (0.003 sec)

mysql>
mysql> INSERT INTO DISPLAY_SLOT VALUES
    -> (1,1,'Main Hall'),
    -> (1,2,'Special Gallery'),
    -> (2,1,'Art Hall'),
    -> (2,2,'Painting Room');
Query OK, 4 rows affected (0.065 sec)
Records: 4  Duplicates: 0  Warnings: 0
```

```
mysql>
mysql> INSERT INTO CONDITION_REPORT
    -> (Loan_ID,Artifact_ID,Report_Date,Condition_Status,Remarks)
    -> VALUES
    -> (1,1,'2026-01-01','Good','No damage'),
    -> (1,2,'2026-01-01','Fair','Minor scratches'),
    -> (2,3,'2026-02-01','Excellent','Good condition'),
    -> (3,5,'2026-03-01','Good','No damage');
Query OK, 4 rows affected (0.070 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql>
mysql>
mysql> -- ARTIFACT_OWNER
Query OK, 0 rows affected (0.003 sec)

mysql>
mysql> INSERT INTO ARTIFACT_OWNER VALUES
    -> (1,3,100,'2025-01-01',NULL),
    -> (2,3,100,'2025-01-01',NULL),
    -> (3,4,100,'2025-01-01',NULL),
    -> (4,4,100,'2025-01-01',NULL),
    -> (5,5,100,'2025-01-01',NULL);
Query OK, 5 rows affected (0.083 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql>
mysql>
mysql> -- LENDER_CONTACT
```

```
mysql>
mysql> -- QUERY 1: DISPLAY ALL ARTIFACTS
Query OK, 0 rows affected (0.002 sec)

mysql>
mysql> SELECT * FROM ARTIFACT;
+-------------+---------------+-----------------+-----------------------+
| Artifact_ID | Artifact_Name | Collection_Type | Description           |
+-------------+---------------+-----------------+-----------------------+
|           1 | Ancient Coin  | Metal           | Old gold coin         |
|           2 | Stone Statue  | Sculpture       | Historical statue     |
|           3 | Old Painting  | Painting        | Traditional painting  |
|           4 | Bronze Pot    | Metal           | Ancient bronze pot    |
|           5 | Wooden Box    | Wood            | Historical wooden box |
+-------------+---------------+-----------------+-----------------------+
5 rows in set (0.005 sec)

mysql>
mysql>
mysql> -- QUERY 2: DISPLAY LOAN AND LENDER DETAILS
Query OK, 0 rows affected (0.002 sec)

mysql>
mysql> SELECT
    -> L.Loan_ID,
    -> P.First_Name,
    -> P.Last_Name,
    -> L.Start_Date,
    -> L.Expiry_Date
    -> FROM LOAN L
    -> JOIN LENDER LE
    -> ON L.Lender_ID = LE.Lender_ID
    -> JOIN PERSON P
    -> ON LE.Lender_ID = P.Person_ID;
+---------+------------+-----------+------------+-------------+
| Loan_ID | First_Name | Last_Name | Start_Date | Expiry_Date |
+---------+------------+-----------+------------+-------------+
|       1 | Ashok      | Kumar     | 2026-01-01 | 2026-12-01  |
|       2 | Vijay      | Kumar     | 2026-02-01 | 2026-10-01  |
|       3 | Mani       | Raja      | 2026-03-01 | 2026-09-01  |
```

```
mysql> -- QUERY 3: FIND OVERDUE LOANS STILL OPEN
Query OK, 0 rows affected (0.001 sec)

mysql>
mysql> SELECT *
    -> FROM LOAN
    -> WHERE Expiry_Date < CURDATE()
    -> AND Return_Date IS NULL;
+---------+-----------+------------+------------+-------------+-------------+
| Loan_ID | Lender_ID | Curator_ID | Start_Date | Expiry_Date | Return_Date |
+---------+-----------+------------+------------+-------------+-------------+
|       2 |         4 |          1 | 2026-02-01 | 2026-10-01  | NULL        |
+---------+-----------+------------+------------+-------------+-------------+
1 row in set (0.012 sec)

mysql>
mysql>
mysql> -- QUERY 4: FIND ARTIFACTS WITHOUT CONDITION REPORTS
Query OK, 0 rows affected (0.001 sec)

mysql>
mysql> SELECT
    -> A.Artifact_ID,
    -> A.Artifact_Name
    -> FROM ARTIFACT A
    -> LEFT JOIN CONDITION_REPORT C
    -> ON A.Artifact_ID = C.Artifact_ID
    -> WHERE C.Report_ID IS NULL;
+-------------+---------------+
| Artifact_ID | Artifact_Name |
+-------------+---------------+
|           4 | Bronze Pot    |
+-------------+---------------+
1 row in set (0.003 sec)

mysql>
mysql>
mysql> -- QUERY 5: DISPLAY ARTIFACTS UNDER EACH LOAN
Query OK, 0 rows affected (0.001 sec)
```

```
ysql>
ysql> SELECT *
   -> FROM ARTIFACT
   -> WHERE Artifact_ID IN (
   ->     SELECT Artifact_ID
   ->     FROM LOAN_OBJECT
   ->     WHERE Loan_Value > 50000
   -> );
-------------+--------------+----------------+---------------------+
 Artifact_ID | Artifact_Name | Collection_Type | Description         |
-------------+--------------+----------------+---------------------+
           2 | Stone Statue  | Sculpture       | Historical statue   |
           3 | Old Painting  | Painting        | Traditional painting |
-------------+--------------+----------------+---------------------+
 rows in set (0.005 sec)

ysql>
ysql>
ysql> -- ========================================
uery OK, 0 rows affected (0.001 sec)

ysql> -- INDEX
uery OK, 0 rows affected (0.001 sec)

ysql> -- ========================================
uery OK, 0 rows affected (0.001 sec)

ysql>
ysql> CREATE INDEX idx_loan_lender
   -> ON LOAN(Lender_ID);
uery OK, 0 rows affected (0.611 sec)
ecords: 0  Duplicates: 0  Warnings: 0

ysql>
ysql> CREATE INDEX idx_condition_artifact
   -> ON CONDITION_REPORT(Artifact_ID);
```

```
mysql>
mysql> -- DISPLAY VIEW
Query OK, 0 rows affected (0.004 sec)

mysql>
mysql> SELECT * FROM LOAN_DETAILS;
+---------+------------+-----------+------------+-------------+
| Loan_ID | First_Name | Last_Name | Start_Date | Expiry_Date |
+---------+------------+-----------+------------+-------------+
|       1 | Ashok      | Kumar     | 2026-01-01 | 2026-12-01  |
|       2 | Vijay      | Kumar     | 2026-02-01 | 2026-10-01  |
|       3 | Mani       | Raja      | 2026-03-01 | 2026-09-01  |
+---------+------------+-----------+------------+-------------+
3 rows in set (0.026 sec)

mysql>
mysql>
mysql> -- ======================================
Query OK, 0 rows affected (0.004 sec)

mysql> -- FINAL DATABASE DISPLAY
Query OK, 0 rows affected (0.003 sec)

mysql> -- ======================================
Query OK, 0 rows affected (0.005 sec)

mysql>
mysql> SHOW TABLES;
+-------------------------+
| Tables_in_meridianmuseum |
+-------------------------+
| artifact                |
| artifact_owner          |
| condition_report        |
| curator                 |
| display_slot            |
| employee                |
| exhibition              |
| lender                  |
| lender_contact          |
| loan                    |
| loan_details            |
```

```
    ->         INSERT INTO PERSON
    ->         (First_Name, Last_Name, Email)
    ->         VALUES
    ->         (
    ->             CONCAT('Person',i),
    ->             'Museum',
    ->             CONCAT('person',i,'@museum.com')
    ->         );
    ->
    ->         INSERT INTO LENDER
    ->         VALUES (LAST_INSERT_ID(),'Individual');
    ->
    ->         SET i = i + 1;
    ->
    ->     END WHILE;
    -> END //
Query OK, 0 rows affected (0.277 sec)

mysql>
mysql> DELIMITER ;
mysql> CALL GENERATE_LENDERS(100);
Query OK, 1 row affected (14.331 sec)

mysql> CALL GENERATE_LENDERS(1000);
ERROR 1062 (23000): Duplicate entry 'person1@museum.com' for key 'person.Email'
mysql> CALL GENERATE_LENDERS(10000);
ERROR 1062 (23000): Duplicate entry 'person1@museum.com' for key 'person.Email'
mysql> CALL GENERATE_LENDERS(100000);
ERROR 1062 (23000): Duplicate entry 'person1@museum.com' for key 'person.Email'
mysql> SELECT COUNT(*) AS Total_Lenders
    -> FROM LENDER;
+---------------+
| Total_Lenders |
+---------------+
|           103 |
+---------------+
1 row in set (0.023 sec)
```